

GOLDEN SECTION SEARCH

Lecture 21: Presented by Jacomine Grobler



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Heuristic

- *To find*
- Rule-based search strategy
- May be applied to non-differentiable functions
- Prone to becoming trapped at local optima

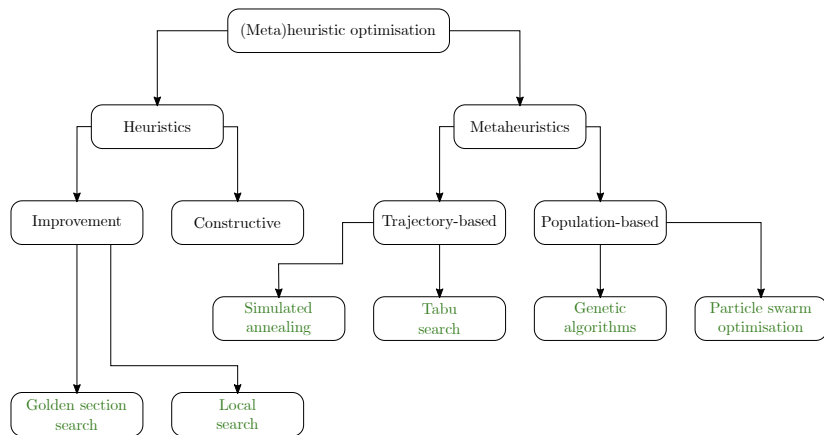
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- *To find*
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Metaheuristic

- *After or beyond*
- Higher-order search strategies
- Capable of escaping local optima.

Heuristics and metaheuristics



The golden ratio

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Using the well-known quadratic formula therefore yields the golden ratio

$$\Phi = \frac{\sqrt{5} - 1}{2} \approx 0.618\,034.$$

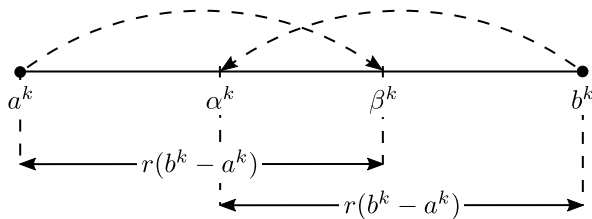
An interval of uncertainty $\mathcal{I}^k = [a^k, b^k]$ (during iteration k) is iteratively reduced in such a manner that if the original interval of uncertainty $\mathcal{I}^0 = [a^0, b^0]$ were to contain a single local optimum of the function f , then so too would all subsequent intervals of uncertainty $\mathcal{I}^k$, for any $k \in \mathbb{N}$.

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The golden section search heuristic therefore returns as output not a single point estimate of a local optimum of the function f , but rather an entire, sufficiently small interval of uncertainty which is guaranteed to contain a local optimum of f .

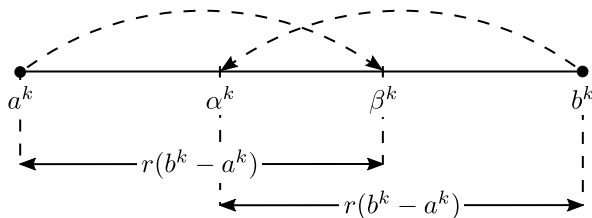
Univariate golden section search

$$\alpha^k = b^k - r(b^k - a^k) \quad \text{and} \quad \beta^k = a^k + r(b^k - a^k)$$



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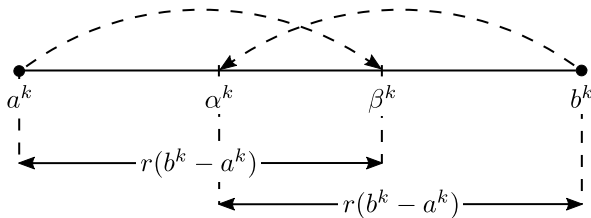
If

$$f(\alpha^k) = \min\{f(\alpha^k), f(\beta^k)\},$$

then the minimum of f in $\mathcal{I}^k$ cannot occur in the portion $[\beta^k, b^k]$ and the next interval of uncertainty is $\mathcal{I}^{k+1} = [a^k, \beta^k]$.

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Similarly, if

$$f(\beta^k) = \min\{f(\alpha^k), f(\beta^k)\},$$

then the next interval of uncertainty is $\mathcal{I}^{k+1} = [\alpha^k, b^k]$.

Univariate golden section search

If, instead of attempting to minimise f , the objective is to maximise a univariate function $f : \mathbb{R} \mapsto \mathbb{R}$ exhibiting a single local maximum in the initial interval of uncertainty $\mathcal{I}^0 = [a^0, b^0]$, then exactly the same observations may be made as described above, except that the minimum operators are replaced by maximum operators.

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The required number of search iterations is

$$\Omega = \left\lceil \log_{\Phi} \frac{\varepsilon_{\text{tol}}}{b^0 - a^0} \right\rceil,$$

where the standard ceiling notion $\lceil x \rceil$ denotes “rounding up of a real number x to the nearest integer” (*i.e.* the smallest integer greater than, or equal to, x) and $\varepsilon_{\text{tol}} > 0$ is the pre-specified uncertainty tolerance value.

Univariate golden section search

Algorithm 4.1 (Golden section search heuristic for univariate functions)

Input: A univariate function $f : \mathbb{R} \mapsto \mathbb{R}$, an initial interval of uncertainty $\mathcal{I}^0 = [a^0, b^0]$ containing a single local optimum of f , and an accuracy tolerance threshold value $\varepsilon_{\text{tol}} > 0$

Output: An interval of uncertainty $\mathcal{I}^\Omega = [a^\Omega, b^\Omega]$ of width at most ε_{tol} containing the original local optimum of the function f in $\mathcal{I}^0$

Step 1 $\Omega \leftarrow \lceil \log_\Phi \varepsilon_{\text{tol}} / (b^0 - a^0) \rceil$

Step 2 For $k = 0$ to $\Omega - 1$ do

Step 3 $\alpha^k \leftarrow b^k - \Phi(b^k - a^k), \beta^k \leftarrow a^k + \Phi(b^k - a^k)$

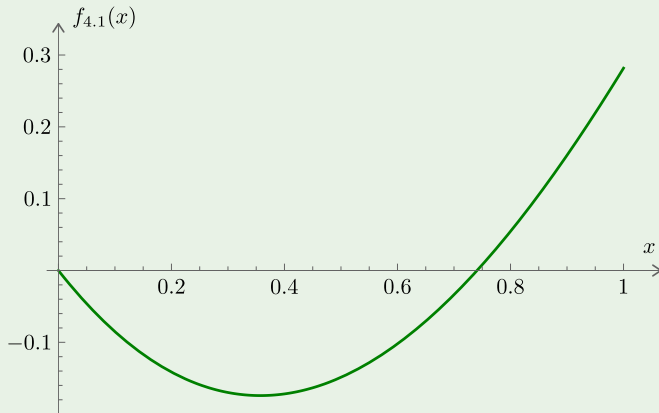
Step 4 If $f(\alpha^k) = \min / \max \{f(\alpha^k), f(\beta^k)\}$, then $a^{k+1} \leftarrow a^k, b^{k+1} \leftarrow \beta^k$

Step 5 else $a^{k+1} \leftarrow \alpha^k, b^{k+1} \leftarrow b^k$

Step 6 Output $[a^\Omega, b^\Omega]$ and stop

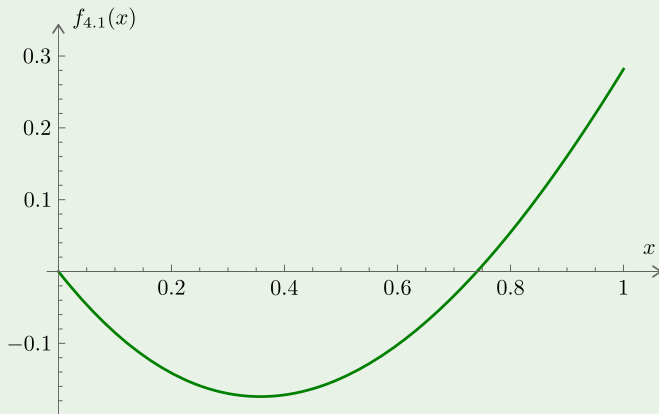
Example 4.1 (pp. 111–113)

Suppose we would like to approximate the local minimum x^* of the function $f_{4.1}(x) = 1 + 2x^2 - e^x$ within the unit interval $[0, 1]$.



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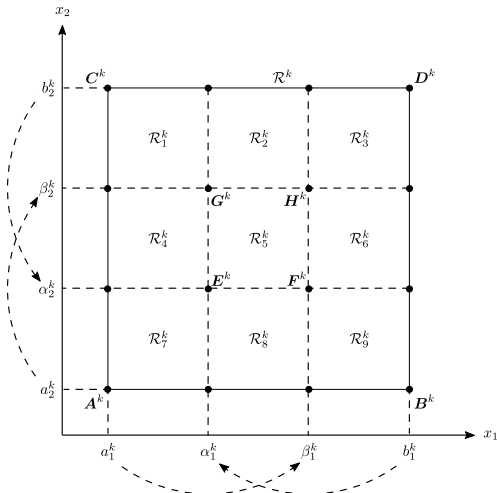
Terminate the search once the length of the interval of uncertainty drops below $\varepsilon_{\text{tol}} = 0.0005$ for the first time. This will require $\Omega = \lceil \log_{\Phi} 0.0005 \rceil = \lceil 15.795337 \rceil = 16$ iterations.

Bivariate golden section search

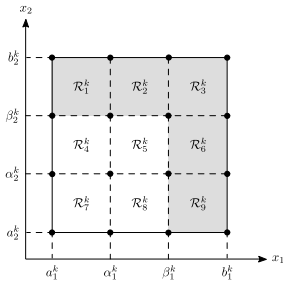
The interval $\mathcal{I}^k = [a^k \ b^k]$ of uncertainty is replaced by that of a *rectangle of uncertainty*, denoted by $\mathcal{R}^k = [a_1^k \ b_1^k] \times [a_2^k \ b_2^k]$, which has corner points $\mathbf{A}^k$, $\mathbf{B}^k$, $\mathbf{C}^k$ and $\mathbf{D}^k$.

Bivariate golden section search

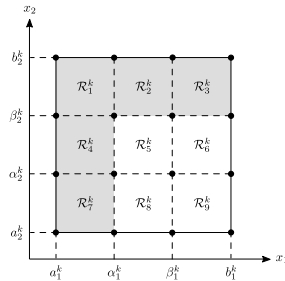
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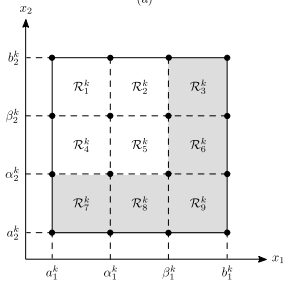
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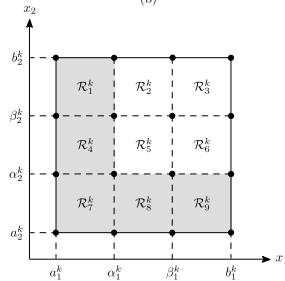
(a)



(b)



(c)



(d)

The required number of search iterations is

$$\Omega = \max \left\{ \left\lceil \log_{\Phi} \frac{\varepsilon_{\text{tol}}}{b_1^0 - a_1^0} \right\rceil, \left\lceil \log_{\Phi} \frac{\varepsilon_{\text{tol}}}{b_2^0 - a_2^0} \right\rceil \right\}.$$

Bivariate golden section search

Algorithm 4.2 (Golden section search heuristic for bivariate functions)

Input: A bivariate function $f : \mathbb{R}^2 \mapsto \mathbb{R}$, an initial rectangle of uncertainty $\mathcal{R}^0 = [a_1^0, b_1^0] \times [a_2^0, b_2^0]$ containing a single local optimum of f , and an accuracy tolerance threshold value ε_{tol}

Output: A rectangle of uncertainty $\mathcal{R}^\Omega = [a_1^\Omega, b_1^\Omega] \times [a_2^\Omega, b_2^\Omega]$ with largest side length at most ε_{tol} containing the original local optimum of the function f in $\mathcal{R}^0$

Step 1 $\Omega \leftarrow \max\{\lceil \log_\Phi \varepsilon_{\text{tol}} / (b_1^0 - a_1^0) \rceil, \lceil \log_\Phi \varepsilon_{\text{tol}} / (b_2^0 - a_2^0) \rceil\}$

Step 2 For $k = 0$ to $\Omega - 1$ do

Step 3 $\alpha_i^k \leftarrow b_i^k - \Phi(b_i^k - a_i^k), \beta_i^k \leftarrow a_i^k + \Phi(b_i^k - a_i^k), i = 1, 2$

Step 4 $\mathbf{E}^k \leftarrow [\alpha_1^k \ \alpha_2^k]^T, \mathbf{F}^k \leftarrow [\beta_1^k \ \alpha_2^k]^T, \mathbf{G}^k \leftarrow [\alpha_1^k \ \beta_2^k]^T, \mathbf{H}^k \leftarrow [\beta_1^k \ \beta_2^k]^T$

Step 5 If $f(\mathbf{E}^k) = \min / \max\{f(\mathbf{E}^k), f(\mathbf{F}^k), f(\mathbf{G}^k), f(\mathbf{H}^k)\}$ then

Step 6 $a_1^{k+1} \leftarrow a_1^k, b_1^{k+1} \leftarrow \beta_1^k, a_2^{k+1} \leftarrow a_2^k, b_2^{k+1} \leftarrow \beta_2^k$

Step 7 Else if $f(\mathbf{F}^k) = \min / \max\{f(\mathbf{E}^k), f(\mathbf{F}^k), f(\mathbf{G}^k), f(\mathbf{H}^k)\}$ then

Step 8 $a_1^{k+1} \leftarrow \alpha_1^k, b_1^{k+1} \leftarrow b_1^k, a_2^{k+1} \leftarrow a_2^k, b_2^{k+1} \leftarrow \beta_2^k$

Step 9 Else if $f(\mathbf{G}^k) = \min / \max\{f(\mathbf{E}^k), f(\mathbf{F}^k), f(\mathbf{G}^k), f(\mathbf{H}^k)\}$ then

Step 10 $a_1^{k+1} \leftarrow a_1^k, b_1^{k+1} \leftarrow \beta_1^k, a_2^{k+1} \leftarrow \alpha_2^k, b_2^{k+1} \leftarrow b_2^k$

Step 11 Else $a_1^{k+1} \leftarrow \alpha_1^k, b_1^{k+1} \leftarrow b_1^k, a_2^{k+1} \leftarrow \alpha_2^k, b_2^{k+1} \leftarrow b_2^k$

Step 12 Output $[a^\Omega, b^\Omega]$ and stop

Example 4.2 (pp. 115–117)

Maximise the function

$$f_{4.2}(x_1, x_2) = -x_1^2 - 4x_2^2 + 2x_1 + 8x_2 - 5$$

approximately by carrying out a golden section search from the initial rectangle of uncertainty $\mathcal{R}^0 = [0, 2] \times [0, \frac{3}{2}]$.

